QUARTERLY JOURNAL
OF THE
ROYAL METEOROLOGICAL SOCIETY

Vol. 96 APRIL 1970 No. 408

Quart. J. R. Met. Soc. (1970), **96**, pp. 157-185

551.511.3 : 551.515

Transfer properties of the large-scale eddies and the general circulation of the atmosphere

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(Manuscript received 7 July 1969; in revised form 2 December 1969)

Summary

The transfer of heat and momentum by motion on the scale of cyclone waves and long waves is deduced from a knowledge of their mechanics. The law of horizontal transfer of entropy is independent of notions of mixing but is similar to that for non-isotropic diffusion. The law accurately represents the annual variation of heat-flux in the Northern Hemisphere. Perturbation theory suggests realistic spatial variations for the transfer and provides a rational basis for calculating transfer coefficients which can then be used to find the flux of other conservative quantities. Most importantly the Equatorward flux of potential vorticity is well defined and is shown to determine the horizontal flux of momentum $\overline{u'v'}$. The mean surface winds, and the mean meridional circulation (which is seen to be frictionally driven) are then determined. A model of the general circulation is defined by these properties, and this is integrated analytically for a simplified case and compared with motion observed in the Gulf Stream, the mesosphere, and dishpan experiments as well as the troposphere.

Summary of less conventional symbols

The overbar with suffix is used to define an average with respect to the suffix symbol.

$\Delta\phi$: difference in ϕ at same height on each side of a baroclinic zone

ΔU : difference in U at top and bottom of layer

$B = \partial\phi_0/\partial z$: a measure of the static stability

ϵ : average slope of isentropic surfaces in troposphere

$\sigma(v)$ or σ_v : standard deviation of v

$\beta = \overline{(\partial f/\partial y)}^{y}$

$\beta_1 = \beta - \overline{(\partial^2 U/\partial y^2)}^{yz}$

$\beta_2 = \beta - \overline{(\partial^2 U_s/\partial y^2)}^{y}$

$\gamma = gB\,H^2\,\beta_1/(f^2\,\overline{\Delta U}^{y})$: roughly the ratio of vorticity change due to meridional advection to that due to stretching

Q : potential vorticity as in Eq. (6)

Q_* : modified potential vorticity as in Eq. (11)

λ : x-wavenumber

$c = c_0 + i\,c_1$: complex wave velocity indicating displacement and amplification.

1. Introduction: the general circulation and the role played by the large-scale eddies

The net radiative heating of the Earth and atmosphere in low latitudes is largely balanced by a polewards and upwards advection of energy by large-scale motion. This motion is essentially turbulent in that the transient velocities are much greater than the mean velocities (at least in the vertical and meridional directions) and the patterns are not to any useful extent repetitive: conservatively the estimate is that there are 10^{25} independent states for the crudest possible picture of the instantaneous global flow. In spite of this complexity there are average quantities which, varying relatively slowly from month to month, define a general circulation. It is with the inter-relation of these quantities that we are concerned here.

One approach to the problem of the general circulation is through numerical models in which the large-scale eddy motion is explicit. These must be capable of representing considerable detail. They must have at least two levels (to indicate the baroclinity) and (say) 5×5 parameters in the horizontal. Integration of this number of simultaneous equations demands the use of a computer and refined numerical techniques (to prevent the numerical instabilities from overpowering the dynamical), and the models can then be integrated for periods of time such that features characteristic of the general circulation are generated. Reassuring though such integrations are in showing that the basic laws of fluid mechanics are not grossly in error and that the required numerical accuracy can be maintained we can, and should, also seek a more understandable theory of the general circulation. The power of the detailed models is not at issue, particularly for weather prediction and as diagnostic tools. But neither is the construction of simplified models a mere academic exercise. For instance the predictions of detailed models should be judged only relative to the predictions of simpler models. Also the economy of a representation in which eddy motion is implicit allows an enormously extended period of integration for the same numerical labour. Perhaps most important is that the effort of constructing a theory of eddy fluxes prompts interesting questions concerning the details of the large-scale eddies themselves.

Some features of the general circulation can be explained simply while others are essentially complicated. In particular the zonal-mean zonal component of the wind and the zonal-mean temperature must satisfy the thermal wind equation quite accurately – thus $\partial u/\partial z$ can be found from $\partial \phi/\partial y$ (where a nearly-cartesian co-ordinate system is used in which x is W-E, y is S-N and z upwards, and u v w are corresponding components of velocity). It seems reasonable to suppose that the temperature distribution could be found from the rate of heating if the mechanism for heat transfer were quantitatively understood. For instance if the transfer were by conduction then the temperature would satisfy a diffusion equation in which the heating enters as a source term, and such an equation, together with suitable boundary conditions, can be integrated to give the temperature. It will be argued later (Sections 3 and 8) that the transfer is by large-scale eddies, resembles non-isotropic conduction, and that this process is sufficiently well understood to allow a useful first approximation to the temperature to be found from the heating.

It is necessary to distinguish rather carefully between the transfer by eddies, and that by the mean circulation. Here we shall define the eddy flow $\mathbf{V}'$ through the relation $\rho\mathbf{V}' = \rho\mathbf{V} - \overline{\rho\mathbf{V}}^x$. It follows that there is no eddy flux of potential energy gz across any latitude circle at any height but only an eddy flux of sensible energy $c_p T$ and latent energy Lr. In contrast the flux of potential energy is comparable with the other two in the mean circulation. However, the most important distinction between the mean and the eddy motion concerns their dynamics. The eddy motion is a convective instability of the meridional gradient of temperature; the intensity of the motion is related directly to the magnitude of the temperature gradient. In contrast the mean meridional circulation is dependent upon frictional interaction of the prevailing winds with the surface : given the three main latitude belts of zonal surface winds (roughly westerlies between 30° and 60° latitude and easterlies elsewhere) frictionally induced flow across the isobars at low levels together with a return flow aloft constitutes the well known 3-cell circulation. The problem is to account for the surface winds.

Unfortunately there are many oversimplified versions of the cause of the mean meridional circulation, usually of the variety in which ' air tends to be thrown back towards the Equator ' or ' gets piled up in the subtropics.' However, Jeffreys (1933) showed that, since the total mass flux across any latitude circle was (observed to be) small the angular momentum carried by the mean circulation must be much less than that removed by friction at the surface. He inferred that if meridional circulation, friction and heating were the only mechanisms operative then the surface wind would vanish practically everywhere. Rossby (1941) essentially reproduced Jeffreys' argument, with some refinement which shows that there would be surface westerlies close to the pole, but the remainder of his argument showing the development of the 3-cell picture is by no means conclusive. In a similar argument it is said that at least the direct tropical (Hadley) cell exists so as to transfer energy polewards in these latitudes. Now the circulation may well do this, yet the evidence for existence is not the same as a cause (see also Lorenz 1967, p. 55). Given the (observed) horizontal temperature gradient the air will not circulate unless the thermal wind balance is destroyed, and the sense of the circulation will be such as to restore balance (at least where the Richardson number is large and positive). The circulation and the heat-flux can persist only if the equilibrium is continuously destroyed by the action of some means of momentum transfer other than the meridional circulation itself. Since the effect of vertical transfer by small-scale eddies in the middle troposphere is probably very much smaller than that of the horizontal transfer by large-scale eddies we are forced to consider the latter as an essential part of the mechanism of the meridional circulation.

2. Properties of the large-scale eddies : theoretical considerations

By considering the momentum budget of latitude belts Jeffreys (1926) showed that the large-scale eddies must be responsible for the maintenance of the surface winds. Charney (1947) and Eady (1949) in their studies of the stability of idealized systems showed that quasi-geostrophic, large-scale turbulence was inevitable under typical atmospheric conditions, and Eady determined the general character of the associated transfer. Most importantly he showed that, since the kinetic energy of the growing eddies is derived from the potential energy of the original flow, the eddies must re-arrange the system towards a state of minimum potential energy so that entropy must be transferred polewards and upwards, which is in the sense required to account for the observed deficit of radiative energy. It is, however, rather more difficult to account for the momentum transfer.

If the unperturbed wind is assumed independent of latitude (as in the theories of Eady and Charney) then the momentum transfer due to the small amplitude motion is also independent of latitude (there is a small non-linear effect, (White 1968), but this is feeble compared to the one sought and as it vanishes when integrated over depth it cannot contribute to maintenance of surface winds). In order to find realistic momentum transfer it is necessary to discuss the stability of flows which vary with latitude as well as with height. If the unperturbed flow varies with latitude only then no potential energy is available for conversion, the eddy kinetic energy must be derived from the kinetic energy of the unperturbed flow and the growing eddies must transfer momentum down the mean gradient of momentum. This is almost exactly in the opposite direction to that needed to maintain the observed surface winds and in addition there is no horizontal transfer of sensible energy. On these, as well as on more general grounds, we conclude that this (barotropic) type of eddy motion does not play an important role, at least in the troposphere.

There have been several studies of the stability of flow in which the vertical shear varies with latitude and amplifying waves capable of transferring westerly momentum towards middle latitudes are found. It is readily seen that the sign of the Reynolds stress $\overline{u' v'}$ depends only upon the latitudinal variation in the longitude of trough and ridge lines and that where these are bowed towards the east there is a convergence of westerly momentum. Fig. 1 shows the result of some calculations by Eady (previously unpublished) in which the origin $Y = 0$ is in the centre of a baroclinic zone extending roughly to $Y = \pm 1$. The troughs and ridges are bowed towards the east nearly everywhere but the Reynolds' stress, which is proportional to A^2, is particularly large at the upper levels, as is observed. Such stability analyses are complicated and it is not easy to generalize about the momentum transfer, though it is evident from similar calculations that the momentum transfer is very much less in the absence of a realistic β-effect. Moreover, as the solutions are strictly valid only while the growing waves are of small amplitude the application

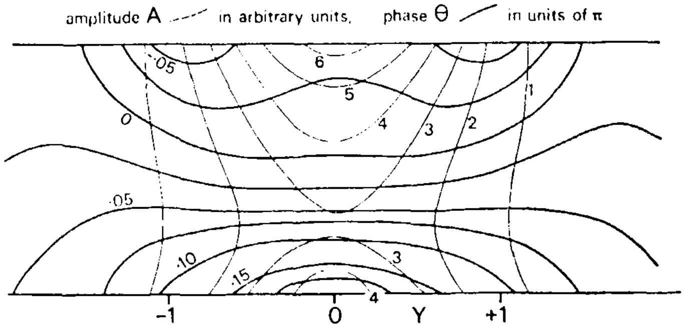

Figure 1. Latitude-height cross-section showing amplitude and phase for a wave growing on a two-dimensional jet-like flow $U(y, z)$ such that $\partial\phi/\partial y$ at $Y = 1$ is half that at $Y = 0$. The β-effect is realistic and corresponds to $\gamma = 1{\cdot}5$, where $\gamma = gBH^2\,\beta_1/f^2\,\overline{\Delta U}^{y}$ and $\beta_1 = \beta - \overline{(\partial^2 U/\partial y^2)}^{y}$. Notice that the phase of the trough and ridge lines is displaced towards the west with increasing distance from the jet, showing an eddy-transfer of momentum towards the jet. From an unpublished calculation by Eady in 1958.

of their results to the behaviour of finite amplitude disturbances is open to some criticism. Thus in this work, and particularly in Section 8 (b), we generalize such results to finite amplitude cases rather cautiously – having regard to more general constraints.

It is worth noting what seems to be at first sight an alternative description of the momentum transfer process (used by Eady as a class example). If initially we have an atmosphere with constant zonal wind in which positive and negative vorticity anomalies have been concentrated in a mid-latitude region then it can be shown that, as these disperse (due to the β-effect) the trough lines become bowed towards the east as required. While these two descriptions apparently differ (one seems to be essentially barotropic, the other essentially baroclinic) both include some representation of the baroclinic process, explicit in the first, implicit (in the origin of the vorticity anomalies) in the second, together with some dispersion which is here due to the variation of Coriolis parameter with latitude. Whereas in the stability theory both effects are considered at the same time, in the barotropic theory they are considered sequentially.

In the present study we extend the theory of amplifying dispersive baroclinic waves by representing their behaviour in terms of a suitably defined transfer theory. The chief characteristic of this theory is that the notion of 'mixing,' with its attendant mystique, is avoided by finding the transfer of quantities, conserved during the eddy-lifetime, directly in terms of trajectories. By this means we relate the transfer of entropy to the mean gradients (Sections 3 and 4) and formalize this transfer in terms of transfer coefficients (Section 8) which represent in essence the distribution of trajectories. These enable the transfer of other conservative quantities to be written down, and directing attention to that of potential vorticity we arrive at an expression for momentum transfer in terms of observable mean quantities (Sections 7 and 8 (d)). Eady (1949) showed that, since the large-scale Richardson number is large, the most active baroclinic waves must in a certain sense be quasi-geostrophic and this property plays an essential role, at least so far as the momentum transfer is concerned. This does not, however, imply that this theory is inapplicable in the Tropics though we avoid that question here by using (for the sake of algebraic simplicity) a co-ordinate system that is.

3. Amplitude of the eddy motion and the associated transfer of entropy

Eddy kinetic energy is derived from potential energy released through the redistribution of mass, and by making a realistic model of this process we can estimate the intensity of the eddy motion. Internal friction and diabatic processes have a somewhat longer time-scale than that of the amplifying eddy so these processes can, to a first approximation, be neglected. Atmospheric motion causes comparatively small deviations of pressure p, density ρ, and dry entropy $c_p\,\phi$, from those appropriate to a mean static atmosphere and it is convenient to expand :

$$p = p_0(z) + \delta p \quad \rho = \rho_0(z) + \delta\rho \quad \phi = \phi_0(z) + \delta\phi.$$

Neglecting higher powers of the deviations, the momentum equation becomes

$$\frac{D\mathbf{V}}{Dt} + f\mathbf{k} \wedge \mathbf{V} + \nabla\left(\frac{\delta p}{\rho_0}\right) - g\mathbf{k}\,\delta\phi = 0 \qquad (1)$$

where the multiplier $\partial\phi_0/\partial z$ has been neglected compared to the operator $\partial/\partial z$. The corresponding continuity equation is

$$\operatorname{div}(\rho_0 \mathbf{V}) = 0. \qquad (2)$$

Scalar multiplication of Eq. (1) with $\rho_0 \mathbf{V}$, re-arrangement using Eq. (2) followed by integration with respect to volume over the whole atmosphere gives

$$\frac{d}{dt}\int (\tfrac{1}{2}\mathbf{V}^2 - gz\,\delta\phi)\,\rho_0\,dx\,dy\,dz = 0, \qquad (3)$$

showing that the sum of kinetic and available potential energy integrated over the whole system is constant. The notion of available potential energy is similar to that due to Lorenz (1955) but the expression in terms of height co-ordinates is more convenient for the present development. Note that Eq. (3) is not true of individual parcels in view of the work done by the pressure (as discussed by Green, Ludlam and McIlveen 1966) but that this cancels out when integrated over the whole system. Also that Eq. (3) has the property (not used here) of being valid even if the initial state is statically unstable.

In order to estimate the intensity of eddy motion consider the potential energy released in an idealized rearrangement. Suppose that initially (Fig. 2 (a)) there is a linear variation of ϕ with latitude, $\Delta\phi$ being the total variation, and that finally the system has been adiabatically rearranged to a state of minimum potential energy (Fig. 2 (b)), in which the isentropic surfaces are horizontal. Recognizing that the mass between any two isentropic surfaces is unchanged the final distribution of ϕ can be found quite generally, though here a simplified model is used. Consider a fluid, incompressible in the sense that the variation of ρ_0 (but not that of ϕ_0) can be neglected. Then it is clear that in Fig. 2 the centre of gravity of the unshaded fluid does not change (each isentrope pivots about its centre point) and a simple integration gives the change in the shaded fluid. Assuming realistically that the horizontal variation of ϕ is somewhat less than the vertical variation, substitution in Eq. (3) then shows that the potential energy released is sufficient to give each particle speed $\mathbf{W}$ where

$$\mathbf{W}^2 = g\,(\Delta\phi)^2/(12\,B) \qquad (4)$$

and $B = \partial\phi_0/\partial z$ is the static stability. This changes slightly during the rearrangement due mainly (and only, *in the incompressible system*) to the spreading of fluid with extreme potential temperatures across the boundaries. This results in an increased stability close to the boundaries illustrated by the dashed line in Fig. 2 (c). However, the modification is insignificant and the value of B appearing in Eq. (4) may be taken as a representative

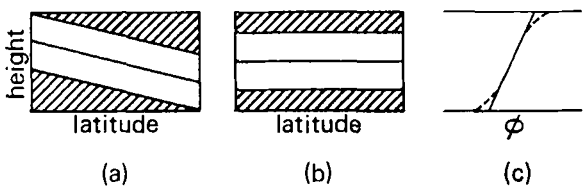

Figure 2. Schematic vertical-meridional cross-section showing isentropic surfaces in an incompressible atmosphere (a) before and (b) after, an adiabatic rearrangement to a state of minimum potential energy. Also (c); the latitudinally averaged variation of ϕ with height before (solid) and after (pecked) the same rearrangement. Notice the spread of very cold fluid over the lower boundary, very warm over the upper – leading to a slight overall increase in static stability.

mean. An important consideration is that no quasi-geostrophic system could, starting from state (a) actually reach state (b). The efficiency of such conversion has been examined by White (1968) who showed that a quasi-geostrophic system is capable of a large fraction of the idealized transition and since this is towards a minimum value of the potential energy the error in Eq. (4) is probably small.

As the potential energy is being released there is also a transfer of entropy and if the meridional component of velocity is supposed proportional to **W** then the northward flux of entropy can be found from

$$\overline{v\phi}^{xtp} = \alpha' \mathbf{W} \, \Delta\phi = \alpha \, (g/B)^{\frac{1}{2}} \, (\Delta\phi)^2 \quad . \quad . \quad . \quad (5)$$

where α' and α are as yet undetermined, dimensionless constants.

In applying Eq. (5) to the atmosphere it is necessary to interpret $\Delta\phi$ in terms of observable quantities. A useful coverage of temperature observations exists between 20° and 70°N and these show, as in Fig. 3, that the mean horizontal temperature difference is roughly constant between 2 and 8 km, above which the gradient changes sign and below which the difference increases because of cold polar surface air, particularly in the winter. Since we are here concerned with tropospheric motion and do not believe that the extreme conditions in the polar surface layer are important we select the 500 mb temperature difference to give a representative value for $\Delta\phi$. This difference is a maximum during the extreme winter months January-February and least during the extreme summer, July-August, so for averages taken over those periods we get an extreme value for $(\Delta\phi)^2$. Taking the winter season, October-March, and summer season, April-September, gives less extreme average temperature differences. It is remarkable that over the same period, while $\Delta\phi$ changes by some 2 : 1 the average static stability in the troposphere changes only by some 4 per cent, so that Eq. (5) predicts that the entropy flux should vary as $(\Delta\phi)^2$.

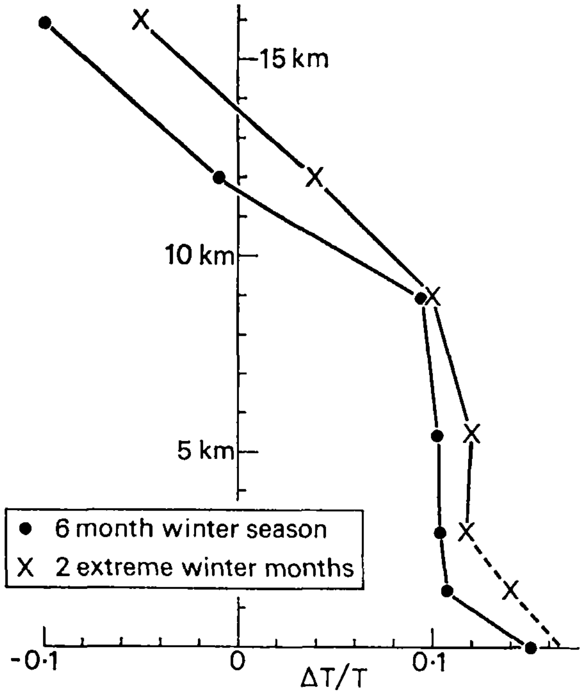

Figure 3. Variation of horizontal temperature gradient with height : fractional difference between temperatures at 20° and 70°N averaged over a period of months :
(•) six winter months of 1950 from Peixoto (1960)
(x) January-February of 1949 from Mintz (1955),
showing nearly constant baroclinity between about 2 and 8 km at each season.

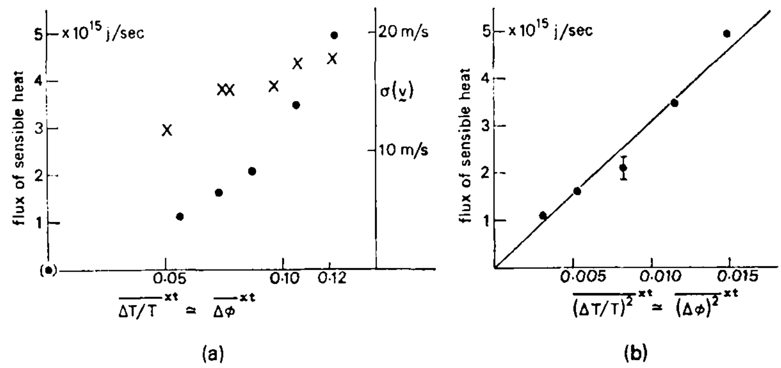

Figure 4. Observed and inferred variation of eddy intensity, and eddy flux of sensible energy, with baroclinity.
(a) (●) Eddy flux of sensible energy E across 50°N : left-hand scale,
(x) mean standard deviation of transient eddy velocity at 500 mb 50°N: right-hand scale, as a function of the latitudinal-temporal average fractional temperature difference at 500 mb : $\overline{(\Delta T/T)}^{xt}$.
(b) E at 50°N plotted against $\overline{(\Delta T/T)^2}^{xt}$.
The vertical line gives the probable error of the flux supposing that the longitudinal variation in the fluxes at the individual stations (about 10 in number) is random. Flux data as in Table 1. Standard deviation vector velocities from Brooks *et al.* (1950) and Buch (1954). According to the baroclinic theory the heat-flux should be linear in (b) and the eddy velocity linear in (a). The heat-flux particularly, shows good agreement with theory.

The entropy flux is nearly proportional to the energy flux and this latter is plotted in Fig. 4 which shows the observed northward eddy flux of sensible heat across latitude 50°N (near the latitude of maximum flux) as a function of $\Delta\phi$, as calculated by various authors. Agreement between theory and observation shown in Fig. 4 (b) must be considered promising.

4. Processing of data

Because the reductions are not entirely objective, the figures, sources and adjustments used in Fig. 4 need to be discussed. Published data on meridional transfer has been analysed in different ways depending mainly upon the definition of 'eddy' chosen by each writer. Two main choices are open – the eddy may be defined as a deviation from temporal-mean flow : $\mathbf{V} - \overline{\mathbf{V}}^t$, or as a deviation from a zonal-mean : $\mathbf{V} - \overline{\mathbf{V}}^x$. If we choose the first definition then transfer due to stationary systems is necessarily excluded, but it seems likely that many of the persistent anomalies in flow are associated with the growth of travelling baroclinic waves in preferred localities – a mere synchronization or triggering of what is essentially a transient phenomenon. For instance it is difficult to escape the impression that the winter 'Icelandic' low is associated with systematic cyclogenesis over the western Atlantic, and is therefore essentially a transient phenomenon. So here we define eddy motion as a deviation from zonal-mean flow, and the transfer can be separated conveniently into the following terms :

$$\overline{vT}^{xt} = E_t + E_s + M_t + M_s$$

where

$E_t = \overline{((v - \bar{v}^x) - (\bar{v}^t - \bar{v}^{xt}))\,((T - \bar{T}^x) - (\bar{T}^t - \bar{T}^{xt}))}^{xt}$, the transient eddy transfer;

$E_s = \overline{(\bar{v}^t - \bar{v}^{xt})\,(\bar{T}^t - \bar{T}^{xt})}^{x}$, the stationary eddy transfer;

$M_t = \overline{(\bar{v}^x - \bar{v}^{xt})\,(\bar{T}^x - \bar{T}^{xt})}^{t}$, the transient mean-meridonal flow transfer;

$M_s = \bar{v}^{xt}\,\bar{T}^{xt}$, the steady mean-meridional flow transfer.

The present theory is intended to apply to the eddy transfer $E = E_s + E_t$ which has been calculated by Mintz (using geostrophic winds) and Starr and White, who also give M_t averaged over one year. Peixoto gives $E_t + M_t$ for the summer and winter seasons, and E_s for the winter season, so to convert his values to $E_s + E_t$ we have assumed that all components of the flux vary proportionally, and used Starr and White to give $M_t = 0{\cdot}18\ E_t$, Peixoto's winter season data to give $E_s = 0{\cdot}32\ E_t$ and adjusted his annual, summer and winter data proportionally. Similarly $\overline{(\Delta T/T)^2}$ has been deduced from $(\overline{\Delta T}/\overline{T})^2$ tabulated by the various authors by multiplying by typical values of $1 + \overline{(\Delta T - \overline{\Delta T})^2}/(\overline{\Delta T})^2$.

TABLE 1. FLUX OF SENSIBLE HEAT DUE TO EDDY MOTION CALCULATED BY THE FOLLOWING AUTHORS: (A) PEIXOTO (1960), (B) STARR AND WHITE (1954), (C) MINTZ (1955), AND ADJUSTED AS IN THE TEXT

The probable error shown in (B) is attributable to the time average. That due to the unrepresentativeness of the space average is $\pm$ 0·28 calculated from Peixoto's data and shown in Fig. 4.

	Season	Flux (10^{15} J s^{-1})		Adjusted E	$\overline{(\Delta T/T)}^{xt}$	$\overline{(\Delta T/T)^2}^{xt}$
A	Annual 1950	$E_t + M_t = 1{\cdot}87$		2·10	0·087	0·0082
	Summer (6 months)	$E_t + M_t = 1{\cdot}46$		1·64	0·073	0·0054
	Winter (6 months)	$E_t + M_t = 3{\cdot}09$	$E_s = 0{\cdot}850$	3·48	0·106	0·0115
B	Annual 1950	$E = 2{\cdot}02$ probable error $\pm$ 0·10	$M_t = 0{\cdot}275$			(0·0082)
C	Jan.-Feb. 1949	$E = 4{\cdot}95$			0·122	0·0150
	July-Aug. 1949	$E = 1{\cdot}12$			0·056	0·0031

5. DETERMINATION OF α FROM FIRST PRINCIPLES

Ideally the value of α would be determined by the theory. It could be found for instance through integration of a detailed numerical model, though this would have few advantages over a direct determination from observation. Here we attempt to indicate the processes determining it and to show that the observed value is consistent in general magnitude with the theory. Supposing the eddy kinetic energy to be divided equally between the zonal and meridional component gives $v \simeq 0{\cdot}7\ \mathbf{W}$, but this is the maximum speed, attained only when all the potential energy available has been released. The average meridional component of eddy velocity is therefore likely to be about half this : $0{\cdot}3\ \mathbf{W}$, giving, for tropospheric values of the other parameters $\sigma(\mathrm{v}) \simeq 100\ \Delta\phi$ m s^{-1}, which is not inconsistent with the slope shown by the data in Fig. 4 (a) : notice however, that the agreement is nothing like so good as for the flux data of Fig. 4 (b).

Similarly, in middle latitudes the extreme variation of ϕ about its mean is $\pm\ \Delta\phi/2$ giving a space-time average variation of about $\Delta\phi/4$, or $\sigma(T) \simeq 6$°C for the troposphere. This is consistent with the general level observed by Peixoto, though the seasonal variation he reports is much smaller than that predicted.

Now the entropy transfer is also proportional to the correlation between v and ϕ which the perturbation theory predicts to be large during the amplifying phase of the eddy (see $r(v\phi)$ in Fig. 6). Non-linear solutions such as that of White (1968) show that it decreases to zero as the eddy reaches maximum amplitude and such a configuration is at least consistent with there being no further release of potential energy. Again, at the surface the coldest air is found at the centre of intense cyclones, not in the southerly stream as it is during their growing phase. The value $\alpha = 0{\cdot}0055$ demanded by Table 1 is obtained if we assume that the correlation between v and ϕ decreases to zero about one quarter of the way through the eddy lifetime and remains zero thereafter. This is consistent with the ratio of amplification to frictional-dissipation time-scales as usually estimated. Since each of these is essentially proportional to the eddy intensity their ratio, hence also α, is expected to be independent of the intensity and therefore of $\Delta\phi$.

The existence of any functional relationship between a mean flux and mean temperature gradient is in contrast to the case of heat transfer due to vertical-overturning convection. There it is clear that the only situations of significance to the heat transfer are convectively unstable. Thus in that case the arithmetical-average temperature, and that relating to the heat transfer process may come from two distinct statistical populations, one including the stable occasions, the other not, and where the convection is intermittent these are likely to be significantly different. This intermittency effect is not present on the large-scale because the flow is always unstable, partly as a consequence of the 'inefficiency' of the transfer.

The data for σ (v) shown in Fig. 4, together with the theory so far presented suggests that there is a considerable background of eddy kinetic energy in addition to that associated with the baroclinic waves. This might arise if there is for example a change in the mean zonal flow whereupon patterns which were previously stationary drift away and are replaced by new stationary patterns. Then the observed kinetic energy will be contaminated (so far as the organized flux-generating eddies are concerned) by disorganized transient motion.

6. Spatial variation of the eddy transfer of entropy

Perturbation solutions describe the shape of eddies at an important stage in their life cycle and we expect this shape to persist, with the modifications suggested above, when the amplitude is no longer infinitesimal. Later, more general though less precise arguments will be developed which account for some of the results of the linearized theory but are no longer dependent on the restrictions of those theories. Initially we concentrate upon a comparison of the observed and linearly-predicted eddy-flux of entropy. The eddy-flux, which is of course a second order quantity, is calculated from the product of two quantities (each of the first order) derived from the linear solutions. Because of our definition of eddy there are no terms of the type 'zero order multiplied by second order' so the first order (linearized) theory gives the eddy-flux correct to one order of smallness. A number of solutions particularly by Eady (1949) and Green (1960) have shown how this depends on parameters of the system and here we summarize some of their results.

(a) *Variation of density with height*

Consider the effect of the general decrease of density with height as it appears in the continuity equation – what has been called the mass-continuity effect because it distinguishes between conservation of mass and volume in the continuity equation. In the expansion of the quasi-geostrophic vorticity equation the effect arises in the vortex-stretching term $f\,\partial\,(\rho w)/\partial z$ where it is apparently important. However, it is found that if the dependent variables are first multiplied by $\rho_0^{\frac{1}{2}}$ the effect disappears almost entirely. Thus $\rho_0^{\frac{1}{2}}\,v$ and $\rho_0^{\frac{1}{2}}\,\delta\phi$ are practically the same in the compressible as in the incompressible system – see e.g. Eady (1949). Accordingly one can ignore the effect of the variation of density with height so long as we compare say $\rho_0\,\overline{v\phi}$ (but not $\overline{v\phi}$) from incompressible theory with observation.

It is not correct to suppose that the effect is eliminated if pressure is used as vertical ordinate, for even though the transformed continuity equation does resemble that for incompressible motion the variation of density reappears in other terms and the effect is still present.

(b) *The variation of Coriolis parameter with latitude*

The 'β-effect' was shown by Rossby (1939) to give barotropic long waves an additive phase velocity towards the west. Charney (1947) showed that this was true of neutral waves in a baroclinic atmosphere and Eady (1949) showed that it was also true of amplifying

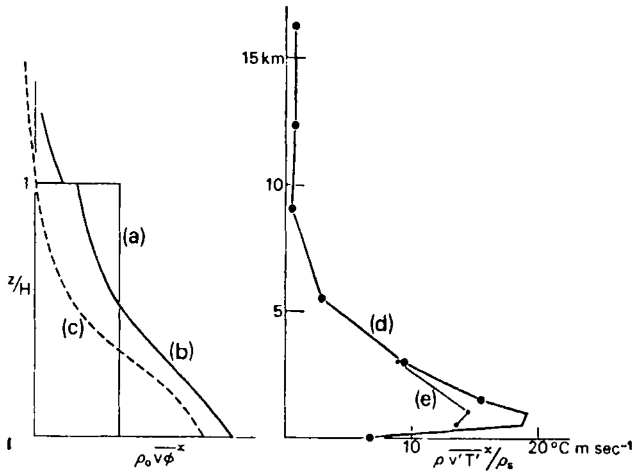

Figure 5. Poleward transfer of entropy by perturbation solutions compared to the observed eddy-flux of sensible energy.

Calculated perturbation of flow with

(a) $\beta = 0$, shear constant (note that the flux is zero above $z = H$).

(b) ' realistic ' β-effect giving $\gamma = 1$, shear constant.

(c) $\beta = 0$ but shear decreases linearly with height to change sign at $z = H$.

Both (a) and (b) have a fourfold increase in static stability at $z = H$.

Observed :

(d) transient eddy flux of energy (practically proportional to entropy) at standard levels by Peixoto (1960) (●).

(e) and extra levels, for a different occasion by Priestley (1949) (•) superimposed to suggest more detailed behaviour in the lower levels.

For comparison of observed data and theory see Section 6.

waves in a baroclinic atmosphere. Now consider waves which amplify in a layer of constant shear bounded above and below by rigid horizontal surfaces. If $\beta = 0$, the equations are symmetrical about the mid-level which is also the steering level (level at which the phase speed is equal to the fluid speed) for the growing waves, and the entropy transfer is independent of height – see Fig. 5 curve (a). If β has a value typical of middle latitudes the phase speed is substantially retarded and (in a westerly shear) the steering level is depressed. Most important for the present purpose, however, the entropy flux is heavily biased towards the surface, as in curve (b) of Fig. 5. A more direct connection between the asymmetry of entropy flux, and the displacement of the steering level will be discussed in Section 8 (b).

(c) *Superimposed layer of greater static stability*

The finite increase in static stability at the tropopause behaves as an imperfect lid to the motion below and as a consequence eddy kinetic energy decreases through the lower stratosphere, with an exponential scale height, typically about 5 km. Vertical motion in the upper troposphere experiences less constraint, and the steering level is raised compared to the situation with a more rigid lid. Typically, however, the effect of β in depressing the steering level dominates over that of the stratosphere in raising it and similarly so far as the distribution of entropy flux is concerned. In Fig. 5, curve (b) shows the asymmetry in the entropy flux when there is a change of static stability of 4 : 1 at $z/H = 1$ but no change of shear in the unperturbed wind.

(d) *Variation of baroclinity with height*

Typically the latitudinal temperature gradient varies considerably with depth through the troposphere and lower stratosphere. Some quite complicated effects of such variation have been discussed, the most interesting for the present purpose occurring when the temperature gradient changes sign. We make use of calculations in which the shear was assumed to decrease linearly with height, to change sign at a level we identify with the tropopause. The static stability was assumed to be independent of height but the solutions are found to have comparatively small amplitude at the tropopause level so the inclusion of a stable stratosphere is unlikely to have a substantial effect. There is no β-effect. The horizontal flux of entropy, shown in curve (c) in Fig. 5, is found to decrease upwards to change sign with the horizontal temperature gradient. However, it is clear that, if there had been a deep layer of constant shear as suggested by Fig. 3 then the entropy flux in the 'troposphere' would have been closer to that given by (a) in Fig. 5, i.e. nearly independent of height.

(e) *Comparison between observation and theory : mid-latitude troposphere*

Condensing the extensive theoretical calculations still further it is evident that the β-effect causes a substantial variation of entropy flux with height but that typical changes of shear and static stability are effective only rather close to the tropopause. In Fig. 5 the distribution of $\rho_0\, \overline{v\phi}^x$ deduced from the linearized theory is compared with representative middle-latitude values of $\rho\, \overline{v' T'}^x$ – technically it would be better to compare $\rho\, \overline{v' T'}^x/T^x$ but the difference is hardly significant and the energy flux is the quantity usually observed. Curve (b) in Fig. 5, which includes the β-effect, is much closer to (d) than (a) which does not, and the modification introduced by allowing the shear to vanish at the tropopause level makes the agreement a little better by decreasing the upper level flux. As stressed in Section 6 (d) the velocity profile used to obtain curve (c) is not realistic in the middle and lower troposphere and the predicted flux would be much more nearly constant with height in those regions if a more realistic profile were used (but keeping $\beta = 0$).

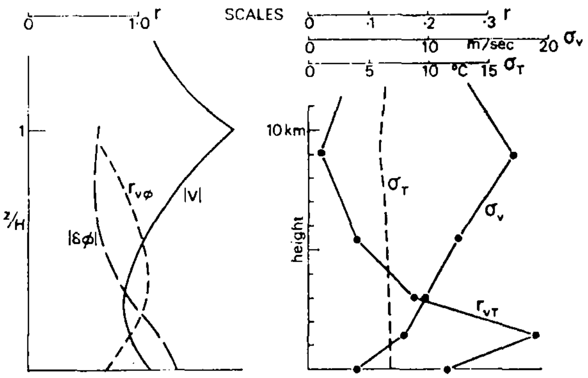

Figure 6. Further breakdown of the data of Fig. 5.

σ_v : standard deviation of meridional component of transient eddy velocity
$|v|$: amplitude of perturbation velocity
σ_T : standard deviation of transient eddy temperature
$|\delta\phi|$: amplitude of perturbation entropy
r_{vT} : correlation between v and T
$r_{v\phi}$: theoretical correlation between v and ϕ (cosine of phase difference).

For comparison of observed data and theory see Section 6 (e).

While the general trend of the variation with height is predicted quite well the linearized theory does not predict the decrease of flux in the surface layers which is probably due to the action of small-scale motion. A more detailed analysis of the flux due to transient eddies at one station by Priestley (1949) shows that the effect is confined to the first kilometre or so (curve (e), Fig. 5). This suggests that it may be due to surface friction. Unfortunately, the data has been analysed with respect to levels of constant pressure, whose height varies from day to day through the depth of the friction layer. An analysis with respect to height would have allowed a more precise interpretation but the difference is likely to be small. Now the perturbation analysis includes friction only through a boundary condition which allows for convergence in a surface friction layer below the lowest level of the calculations. Thus the effect of friction on the motion above the friction layer is included, but the effect on the profiles of wind within the friction layer itself is not. A similar effect concerning the convective transfer of heat from the surface must be expected to modify the temperature similarly : there is a tendency for convective heat transfer to warm the cold air moving south, but not to cool the warm air going north and so give a reduced heat-flux ($\simeq \frac{1}{2}$) in the convective layer. Either or both of these effects can be expected to account for the decreased flux in the surface layer.

Fig. 6 shows a breakdown of Peixoto's data compared with a similar breakdown of the theoretical solution (b) in Fig. 5. The observed standard deviation $\sigma(v)$ compares well with the amplitude of the eddy motion $|\mathbf{V}|$ at all levels except the surface. The discrepancy in the surface layer may well be due to friction. Agreement between $\sigma(T)$ and the corresponding theoretical quantity $|\delta\phi|$ is not so clear but $|\delta\phi|$ has comparatively large values in the lower layers and these would tend to be reduced through convective interaction with the surface.

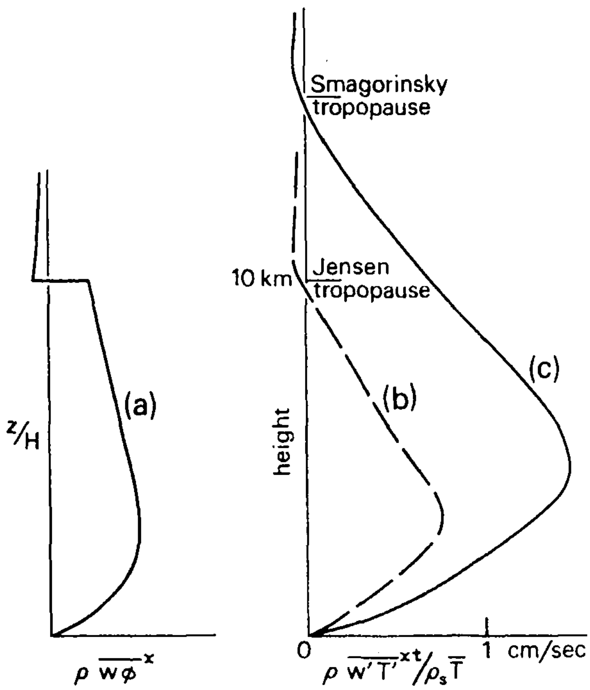

Figure 7. Vertical transfer of entropy by a perturbation solution compared to the observed eddy flux of sensible energy

(a) perturbation of flow with constant shear, increased static stability above $z = H$ and realistic β-effect.
(b) from a N.W.P. scheme using current synoptic data – Jensen (1960)
(c) from a dry, 9-level general circulation model – Smagorinsky (1965), both these latter at about 40°N where the flux is largest.

For completeness the correlations are also shown. The theoretical solution gives high correlation between v and ϕ, but since this covers only the amplifying phase of the motion this is not unexpected. The sharp peak in the observed correlation near 850 mb is not explained by any known perturbation solution and its cause is a matter of some interest. Since the phenomenon is evidently confined to low levels, it may be because the conversion of latent to sensible heat occurs principally in frontal zones where vertical and meridional displacements are highly correlated. In common with several other questions raised in this study it would be extremely useful to be able to refer to the temporal evolution of spatial statistics (like $\rho\ \overline{v\phi}$ say) for those numerical solutions in which the eddies are synchronized.

(f) Observed and linearly-predicted vertical transfer of entropy

Vertical velocities are not directly observed and must be inferred from other quantities after more or less prolonged and uncertain analysis. Jensen (1960) used values computed in a numerical weather prediction scheme and deduced the data shown in Fig. 7, whereas Smagorinsky's data comes from his numerical model working in its general circulation mode i.e. without ' observed ' initial data. These calculations, together with the theoretical results (again the distinction between $\overline{w'T'}$ and $\overline{w\phi}$ has been ignored) show an upward flux in the troposphere with a low level maximum, and a downward flux in the stratosphere. The low level of the maximum in the linearized solutions is again due to the β-effect, and with $\beta = 0$ the maximum would rise to above the middle level, associated with the lesser constraint of the upper boundary compared to the lower.

(g) Entropy transfer in the lower stratosphere : a disagreement with the theory

Though the perturbation theory gives quite realistic profiles for $\overline{v\phi}$ and $\overline{w\phi}$ in the troposphere of middle and high latitudes the comparison is not so good in the lower stratosphere. In the troposphere the warmest air is also ascending most rapidly – temperature and vertical velocity are well correlated – but in the lower stratosphere the sudden increase in static stability forces a high correlation between temperature and vertical displacement, so that the correlation between temperature and vertical velocity is small. Consistently the perturbation theory gives very little (but downward) flux above the tropopause as illustrated by Fig. 7 curve (a).

However, the perturbation theory gives a down-gradient horizontal flux of entropy, whether the horizontal temperature gradient changes sign (as in curve (c) in Fig. 5) or not (as in curve (b) in Fig. 5), whereas a northward flux of heat is observed in the region of northward temperature gradient above about 12 km as illustrated in Figs. 3 and 5.

Now as a consequence of geostrophic and hydrostatic balance it follows that there is an Equatorward transfer of sensible energy so long as the troughlines slope towards the east with height, and some synoptic charts do show such structure in the stratosphere above growing waves. This suggests that the perturbation prediction may be correct for the amplifying phase but that reverse transfer occurs in the mature phase. Smagorinsky, Manabe and Holloway (1965), in their NWP model, which explicitly includes stratospheric motion, find a poleward eddy flux of heat in middle latitudes but the temperature gradient is Equatorward or at least very weak at nearly all heights (Fig. 4 B1, page 736) so their results do not seriously disagree with those of the perturbation theory. With the latent heat of condensation included their computations do produce a jet maximum and reversed temperature gradient but these are weak. On the other hand, Manabe and Hunt (1968) do produce a realistic jet in a similar model which has twice as many gridpoints in the vertical so as to produce resolution of about 3 km.

It seems clear that there is some unexplored difficulty in explaining the reversal of horizontal temperature gradient above the tropopause, so also the upper limit to the tropospheric jet. Heating of the polar regions through ozone absorption in the ultraviolet is insufficient in magnitude and too high in altitude (see for instance Manabe and Strickler (1964) for radiative equilibrium calculations) and the explanation is likely to depend upon an observed, but not satisfactorily explained counter-gradient eddy flux of heat.

7. The eddy transfer of potential vorticity and momentum

In Section 8 we will generalize the theory relating to the eddy transfer of entropy to include the eddy transfer of any conservative quantity, but there is a horizontal flux of momentum defined by the Reynolds' stress $\overline{u'v'}$ which, as argued by Eady (1950) and mentioned in the Introduction, plays an important role in the maintenance of the surface winds. Momentum is not a conservative quantity so its flux cannot be found directly from the transfer theory but potential vorticity is conserved (in frictionless adiabatic flow) so the transfer theory can be used to find its flux. We shall now demonstrate that the flux of potential vorticity is related to that of absolute vorticity and hence to the momentum flux. The required expression has already been derived by Charney and Stern (1962) and Bretherton (1966) in connection with criteria for stability, but here we derive the relation from first principles and consider a more fundamental interpretation.

(a) *The flux of potential vorticity due to quasi-geostrophic eddies*

Potential vorticity Q is defined by Ertel (1942) through

$$\rho Q = (\mathrm{curl}\,\mathbf{V} + 2\boldsymbol{\omega}) \,.\, \nabla\phi \qquad (6)$$

where the notation is conventional, and it can readily be shown that Q is conserved i.e. $DQ/Dt = 0$ for any frictionless adiabatic motion. Here we are concerned with large-scale eddies (zonal wavenumber less than ten or so) which are observed to dominate the meridional transfer of heat and momentum, and Eq. (6) can be simplified considerably by using approximations consistent with the usual quasi-geostrophic model of large-scale flow. These approximations depend mainly upon the largeness of the Richardson number but also on the relative smallness of deviations of pressure and density from those in a mean standard atmosphere. It can be shown that if

$$1 \gg |\zeta/f| \gg (\partial V/\partial z)^2/(gB), \quad |\partial(\delta\phi)/\partial z| \ll B \quad \text{and} \quad |\delta\rho/\rho_0| \ll 1$$

then $\quad \rho\, Q \simeq \rho_0\, Q$

and $\quad \rho_0\, Q \simeq (\zeta + f)\,\partial\phi/\partial z \simeq B\zeta + f\,\partial\phi/\partial z \qquad (7)$

For the most important range of wavenumbers (between about 3 and 6) Eq. (7) is accurate to a little better than 1 per cent of its mean value and a little better than 10 per cent of its variation about the mean, which is the more significant comparison.

Let v be the northward component of eddy velocity, which by definition satisfies $\bar{v}^x = 0$. Multiplying Eq. (7) by v and integrating round a latitude circle gives the northward eddy flux of potential vorticity, but the eddy motion must be nearly geostrophic (because of the largeness of the Richardson number which determines the length scale of the eddy, hence the Rossby number) and the terms may be simplified. From Eq. (7)

$$\rho_0\, \overline{vQ}^x \simeq B\, \overline{v\zeta}^x + f\,(\overline{v\,\partial\phi/\partial z})^x \qquad (8)$$

but $\partial u/\partial x \simeq -\partial v/\partial y$ since the flow is nearly non-divergent, so

$$v\zeta \simeq v\frac{\partial v}{\partial x} - u\frac{\partial u}{\partial x} - \frac{\partial}{\partial y}(uv)$$

and integrating round a latitude circle the first two terms on the right-hand side vanish identically giving

$$\overline{v\zeta}^x \simeq -\,\partial(\overline{uv}^x)/\partial y$$

thus in Eq. (8) the first term represents the divergence of the eddy flux of momentum. In the remaining term on the right-hand side of Eq. (8)

$$\overline{v\,\partial\phi/\partial z}^x = \partial(\overline{v\phi}^x)/\partial z - \overline{\phi\,\partial v/\partial z}^x \simeq \partial(\overline{v\phi}^x)/\partial z$$

because the wind and temperature are in thermal wind balance.

Eq. (8) therefore reduces to

$$\rho_0 \overline{vQ}^x \simeq f \frac{\partial}{\partial z} (\overline{v\phi}^x) - B \frac{\partial}{\partial y} (\overline{uv}^x). \qquad (9)$$

This important equation shows a connection between the eddy fluxes of potential vorticity, potential temperature and momentum. Since two of these are to a first approximation conserved during the eddy motion all three can be treated in terms of a transfer theory. Eq. (9) can also be derived by integrating the equation after 3.12 in Charney and Stern (1962) but the derivation given here is more direct.

Values representative of the troposphere have been extracted from data analysed at MIT by Starr and White and at UCLA by Mintz and their groups :

$$f \,\partial\, (\overline{v\phi}^x)/\partial z \sim -\, 6 \times 10^{-10}\, \text{sec}^{-2} \text{ above about 2 km.}$$

$$B \,\partial\, (\overline{uv}^x)/\partial y \sim \mp\, 2 \times 10^{-10}\, \text{sec}^{-2} \text{ near tropopause, latitude} \gtrless 30^\circ.$$

That the second term is appreciably smaller than the first calls for some comment because it tends to detract from the usefulness of Eq. (9) : evidently the momentum flux sought is a rather small residual between two large terms (but note its inevitability in Section 9 (b)). Moreover it is surprising because when Eq. (7) is applied to the explicit local eddy motion the corresponding terms are strictly comparable. Thus comparing $B\zeta$ with $f\, \partial\, (\delta\phi)/\partial z$ we have

$$\left| \frac{B\zeta}{f\, \partial\, (\delta\phi)/\partial z} \right| \simeq \left| \frac{\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2}{(f^2/gB)\, \partial^2 v/\partial z^2} \right| \qquad (10)$$

and in the problem treated by Eady (1949), for example, this quantity is exactly unity (page 47, Eq. (6)). Moreover, the correlation between the velocity components is roughly the same (*ca* 0·14) as that between temperature and meridional velocity.

It appears that the zonal averaging causes this change of emphasis : in the theory the baroclinic zone is supposed initially to run W-E but in reality it meanders over a range of latitudes. Thus the latitudinal scale of the zonal average is a combination of the eddy scale (given by Eq. (10)) and the latitudinal distance over which the baroclinic zone meanders; since the latter is considerable the latitudinal variations are substantially smoothed.

(*b*) *Flux of Q_* due to quasi-geostrophic eddies*

An even more useful expression results after multiplying Eq. (9) by ρ_0 and rearranging to give

$$\frac{\partial}{\partial y} (\rho_0\, \overline{uv}^x) = \frac{f}{B} \frac{\partial}{\partial z} (\rho_0\, \overline{v\phi}^x) - \frac{\rho_0}{B} \rho_0\, \overline{vQ_*}^x \qquad (11)$$

where

$$\rho_0\, Q_* = \rho_0\, Q + \frac{f}{\rho_0} \frac{\partial \rho_0}{\partial z} \phi = f \frac{\partial \phi}{\partial z} + B\zeta + \frac{f}{\rho_0} \frac{\partial \rho_0}{\partial z} \phi.$$

This is similar to Eq. (9) except that Q has been replaced by $\rho_0\, Q_*$, a quantity introduced into quasi-geostrophic theory because it is conserved during adiabatic, frictionless advection by the horizontal component of the motion – it arises during the elimination of vertical velocity between $D\phi/Dt = 0$ and $DQ/Dt = 0$. Eq. (11) has two main advantages over Eq. (9) : there is no vertical advection of $\rho_0\, Q_*$ so the complication due to the slantwise nature of the motion largely disappears so far as that term is concerned, and Eq. (11) can rather readily be integrated with respect to height to give, on the left-hand side, the contribution of the large-scale momentum transfer to the surface stress (as in Eq. 20).

8. Transfer coefficients and their spatial variation

Classically it has been supposed that there exists a displacement $\mathbf{l}$ say, over which a fluid particle carries the value of the conservative property that it had initially. Thereafter a catastrophic process is imagined which results in the mixing of this particle with its new surroundings. If the particle was initially typical of its surroundings then the flux of the conservative property S is given, after suitable definition of the averaging process by

$$\mathbf{V}S = -\overline{\mathbf{V}\mathbf{l}} \cdot \nabla S$$

where ∇S is assumed to vary little over distances comparable with $\mathbf{l}$. The tensor array $\overline{\mathbf{V}\mathbf{l}}$ then defines coefficients of eddy diffusivity. Unfortunately the displacement $\mathbf{l}$ is not well defined, nor are the 'mixing' processes which determine its value, and a particle due to go on a long excursion is hardly likely to be typical of its surroundings. Moreover the theory has been applied to non-conservative quantities (like momentum) for which it is not appropriate. If the formulation is inverted and used to define the eddy diffusivity in terms of measurable fluxes and gradients then the coefficients are no longer independent of the property S and cannot be predicted. An important achievement of the perturbation theory was to show that for realistic processes the transfers depend on subtle constraints which cannot in general be represented by a crude diffusion theory. This is particularly true of the momentum transfer incurred during convective overturning. Apart from this negative aspect, the perturbation theory indicates the essential structure of the eddies and this more positive aspect is explored here.

Consider the displacements produced by an eddy during its lifetime. Initially all particles are representative of their surroundings and while the amplitude is infinitesimal perturbation theory can be used to find their displacement. Thereafter the theory must be more speculative, and as discussed in Section 3 the overall magnitudes will finally be found from the energetics. Having specified the effect of one eddy it is necessary to consider the superposition of succeeding systems – the analogy is with the discussion of one molecular collision generalized to produce a kinetic theory. But whereas the concept of a molecular collision may be rather highly simplified and still be useful, the eddy event is correspondingly complicated. For transfer in the meridional-vertical plane, the array of transfer coefficients is obtained (as before) through a truncated Taylor-series :

$$\left.\begin{aligned} \overline{vS} &= -\overline{v\Delta y}\frac{\partial S}{\partial y} - \overline{v\Delta z}\frac{\partial S}{\partial z} = -K_{vy}\frac{\partial S}{\partial y} - K_{vz}\frac{\partial S}{\partial z} \\ \overline{wS} &= -\overline{w\Delta y}\frac{\partial S}{\partial y} - \overline{w\Delta z}\frac{\partial S}{\partial z} = -K_{wy}\frac{\partial S}{\partial y} - K_{wz}\frac{\partial S}{\partial z} \end{aligned}\right\} \quad . \quad . \quad (12)$$

where all the averages are with respect to x, and Δy, Δz are the displacements since $t = -\infty$ of the particles currently at the point of interest. This equation is of the same form as that for non-isotropic diffusion, but the coefficients have a different interpretation. In common with that process the transfer is not parallel to the gradient ∇S unless $K_{vz} = K_{wy} = 0$ and this is not likely to be true of the baroclinic motion because of its pronounced slantwise-bias. The K's themselves are geometrically related so there may be functional relationships between them – if so they are not obvious. We will show however, that K_{vy} and K_{wz} are positive, at least in the initial stages of amplifying waves, and K_{vz} and K_{wy} are positive in the initial stages of amplifying baroclinic waves – properties that play an important role in the conclusions to be drawn later.

Displacements can be found from the velocity field by integrating the three simultaneous equations,

$$\frac{d}{dt}\Delta x = u, \quad \frac{d}{dt}\Delta y = v, \quad \frac{d}{dt}\Delta z = w \quad . \quad . \quad . \quad . \quad (13)$$

where d/dt is a Lagrangian derivative with the identity of the particle held constant : u, v, w, are to be regarded as functions of time, and particle identity only.

(a) Transfer coefficients for motion of small amplitude

If an initial flow $U(y, z)$ is perturbed by an amplifying eddy for which

$$v = \text{Real}\{\exp i\lambda (x - ct) F(y, z)\}$$

the linearized right-hand side of Eq. (13) may be integrated to give :

$$x = x_0 + Ut + O(v)$$

$$\left.\begin{aligned} &\text{hence} \quad \frac{d}{dt}\Delta y = \text{Real}\{\exp i\lambda (x_0 + (U - c)t) F(y, z)\} + O(v^2) \\ &\text{and} \quad \Delta y = \text{Real}\left\{\frac{F}{i\lambda (U - c)} \exp i\lambda (x_0 + (U - c)t)\right\} + O(v^2) \end{aligned}\right\} \quad . \quad (14)$$

where Real denotes the real part of the complex expression. Multiplying this expression for Δy with v and averaging over x we get :

$$K_{vy} = \overline{v \Delta y}^x = \frac{c_1 |v|^2}{2\lambda |U - c|^2} + O(v^3) \quad \text{where} \quad c = c_0 + ic_1 \quad . \quad (15)$$

Notice that K_{vy} as given by Eq. (15) is proportional to c_1 (so to the amplification rate) and is positive everywhere. Typically the variation of $|U - c|^2$ dominates over that of $|v|^2$ in the vertical and gives a pronounced maximum to K_{vy} close to the steering level, as illustrated in Fig. 8 (a).

The horizontal variation depends mainly on that of the eddy kinetic energy $|v|^2$ which, as illustrated in Fig. 1, decreases substantially away from the baroclinic zone, but also upon that of $|U - c|^2$, showing a steering-level effect. The net result of these two effects is to make K_{vy} relatively constant over a range of middle latitudes but decreasing to become small away from the latitude at which the steering level comes down to the surface. A similar argument applies to K_{wz}, but the algebraic expressions for the two cross-coefficients are complicated and involve the solution to the stability problem through the relation of v to Δz and w to Δy. The expressions are long and not immediately informative

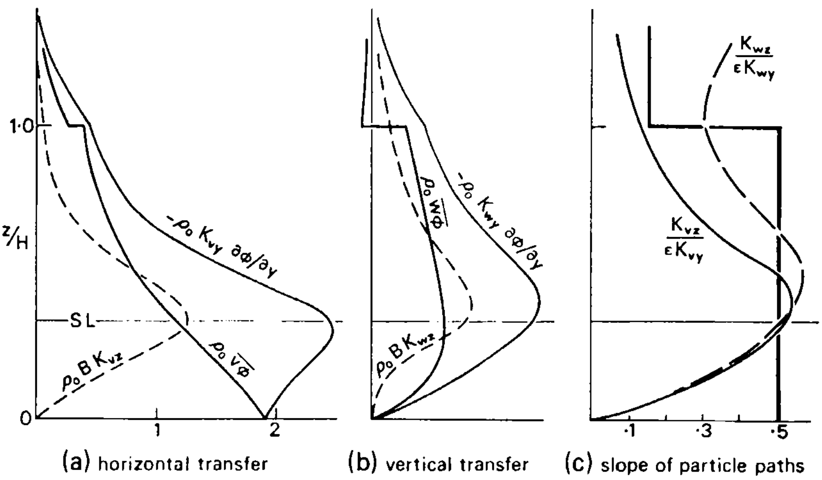

Figure 8. Variation of transfer coefficients with height – constant shear, $\gamma = 1$, calculated from the perturbation solution of Fig. 5 (b).

(a) coefficients for poleward transfer in arbitrary units; Note : $\rho_0 \overline{v\phi}^x = -\rho_0 K_{vy}\, \partial\bar{\phi}/\partial y - \rho_0 B K_{vz}$

(b) coefficients for vertical transfer in arbitrary units; Note : $\rho_0 \overline{w\phi}^x = -\rho_0 K_{wy}\, \partial\bar{\phi}/\partial y - \rho_0 B K_{wz}$

(c) slope of v-weighted, and w-weighted particle paths, showing the close approach to Eady's optimum slope in the troposphere : ϵ is the (constant) slope of isentropic surfaces in the 'troposphere' $0 < z < H$.

so they are not reproduced, but they have been used in conjunction with the 'typical' perturbation solution of (b) in Fig. 5 to calculate the coefficients shown in Fig. 8. These confirm the suspicion that all the K's are positive, at least while the growing wave is of small amplitude and the broad features of their variation can readily be interpreted. However, the chain of reasoning has now become rather long (the literature on the stability problem is extensive) and it is fortunate that one can, with some small sacrifice in precision, discuss the distribution of the K's in a more general manner. This will not be so dependent upon the approximations involved in linearized analyses but can be checked against them.

(b) *Transfer coefficients for motion of finite amplitude*

(i) The displacement Δz satisfies an equation similar to Eq. (14) and, like Δy, includes the factor $1/(U - c)$ which gives the maximum of displacement near the steering level. This maximum displacement occurs because particles move only slowly through, relative to the wave at the steering level (where the unperturbed speed of a particle is equal to the wave speed) so are subject to transverse displacement of the same sense for longer than at any other level. But much the same situation exists even in a mature finite amplitude disturbance, for the relative flow is still mainly zonal and a layer of air gets 'caught up in the wave' and suffers large displacement. This enhanced steering level displacement is an obvious feature of isentropic relative flow charts, and on surface relative flow charts where the steering level comes down to the surface in the warm sector. Thus we expect a maximum value of K near the steering level for disturbances of finite amplitude.

(ii) In addition K_{vz} and K_{wy} are small near the surface, where vertical velocities and displacements are small, and in the stratosphere where vertical movement is inhibited by the stratification. Similarly K_{wz} is small to the second order at the same levels.

(iii) Eady (1949) showed that the orientation of finite displacements needed to give the thermodynamic-maximum rate of release of potential energy (half the slope of the initial isentropic surfaces) was closely approached in the middle levels of his amplifying baroclinic eddies in spite of the dynamic and geometric constraints, and he suggested that this would be true of all such eddy motion – see also Green (1960). An average particle path is not well defined, for the longitudinal-average of the eddy velocities v and w both vanish by definition as do the family of displacements similarly averaged. Similarly Eady's analysis was in terms of parcel interchange, the one replacing the other, so the average of the two displacements again vanishes. But we are not interested in displacements as such, but rather the properties carried by the displaced fluid, in other words correlations between displacements and properties. Thus the average displacement must be associated with some weighting function and Eq. (12) indicates those weighting functions likely to be most important. In Fig. 8 (c) we have plotted the slopes (suitably normalized) of the average particle paths, weighted by v and w, according to the perturbation solution (b) in Fig. 5. If all the displacements had been along the optimum direction suggested by Eady then these slopes would follow the thick lines – which is impossible near the surface and unlikely near the tropopause because of the effect of the boundaries. We conclude that for this perturbation solution the optimum orientation is rather nicely attained near the steering level where transverse displacements are particularly large, but not near the surface or the tropopause where the geometrical constraints dominate and the displacements themselves are relatively small. It is clear that Eady's notion of optimum orientation of finite displacements is a useful guide to the ratio of transfer coefficients near the steering level of a large class of amplifying baroclinic waves, at least while their amplitude is small.

(iv) Since the transfer coefficients depend upon properties of the flow integrated along trajectories, and since particles will in general have traversed a large fraction of the baroclinic zone in their lifetimes, the average coefficient is likely to depend only on properties of the mean flow averaged over the baroclinic zone : for example on the overall baroclinity rather than on an instantaneous local value. Later we make use of this feature by supposing that the K's in a realistic situation are the same as those in a similar system in which the baroclinity is constant : specifically the comparison system will have the same extreme variation of potential temperature but with constant horizontal and vertical gradients, and the same average latitudinal variation of absolute vorticity. This leads to an important relation to be discussed in Section 8 (e).

(c) *The entropy flux expressed in terms of transfer coefficients*

Entropy is closely conserved by the large-scale eddy motion, and putting $S = \phi$ in Eq. (12) gives expressions for the horizontal and vertical eddy flux of entropy

$$\left.\begin{aligned} \overline{v\phi}^x &= - K_{vy}\,\partial\phi/\partial y - K_{vz}\,\partial\phi/\partial z \\ \overline{w\phi}^x &= - K_{wy}\,\partial\phi/\partial y - K_{wz}\,\partial\phi/\partial z. \end{aligned}\right\} \qquad (16)$$

Let $\epsilon = -(\partial\phi/\partial y)/(\partial\phi/\partial z)$ be the initial slope of the isentropic surfaces, then where the displacements are in the optimum direction

$$w = \tfrac{1}{2}\epsilon\, v, \quad \Delta z = \tfrac{1}{2}\epsilon\,\Delta y \quad \text{so} \quad K_{vz} = \tfrac{1}{2}\epsilon\, K_{vy}, \quad K_{wz} = \tfrac{1}{2}\epsilon\, K_{wy} = \tfrac{1}{4}\epsilon^2\, K_{vy}$$

suggesting that, near the steering level all the transfer coefficients should be directly related to K_{vy}. These relations imply

$$B\,K_{vz} = -\tfrac{1}{2}K_{vy}\frac{\partial\phi}{\partial y} \quad \text{and} \quad B\,K_{wz} = -\tfrac{1}{2}K_{wy}\frac{\partial\phi}{\partial y}$$

near the steering level. This is readily seen to be true of the data shown in Fig. 8 which is the result of a perturbation calculation in which all geometrical and dynamical constraints have been satisfied. Moreover, the fluxes near the steering level can be found and similarly related to K_{vy}. It is interesting to notice that vertical motion causes the terms on the right-hand side of Eq. (16) to half cancel near the steering level while (particularly in respect of $\overline{v\phi}$) this compensatory vertical motion must be small near the boundary. This is why, in spite of the motion being in a sense more organized near the steering level, the horizontal flux of entropy is a maximum at the surface and not at the steering level.

We conclude that the condition of optimization near the steering level, together with reasonable profiles of the transfer coefficients, allows them to be related quantitatively to the transfer law of Eq. (5), and in turn to the constant α of that equation. For example, assuming that the vertical distributions of Fig. 8 are representative, then integrating them over the troposphere $0 \leqslant z \leqslant H$ we get

$$\int_0^H \rho_0\, K_{vy}\frac{\partial\phi}{\partial y}\,dz \simeq -1\cdot 5\int_0^H \rho_0\,\overline{v\phi}^{xt}\,dz$$

and using Eq. (5) :

$$\overline{K_{vy}}^{xpt} \simeq -1\cdot 5\,\alpha\,(g/B)^{\frac{1}{2}}\,(\Delta\phi)^2/\overline{(\partial\phi/\partial y)}^{xpt} \qquad (17)$$

where $\Delta\phi$ is the typical (500 mb) mean value, not the local function of height. Assuming that $\overline{\partial\phi/\partial y} \simeq \Delta\phi/a$ where a is the Earth's radius and substituting realistic and uncritical values for B (and g) gives :

$$\overline{K_{vy}}^{xpt} \simeq 0\cdot 8\times 10^{10}\,\alpha\,\Delta\phi \simeq 4\cdot 4\times 10^7\,\Delta\phi\ \mathrm{m^2\,sec^{-1}}\ \text{with}\ \alpha = 0\cdot 55\times 10^{-2}.$$

For the range of temperature differences covered in Table 1 this gives a range of transfer coefficients between 5 and $3\times 10^6\ \mathrm{m^2\,s^{-1}}$ similar to the range of values quoted for large-scale diffusivities by various authors.

An alternative (though not independent) derivation of Eq. (17) is of some interest. Integrating Eq. (15) with respect to z we can take advantage of the relatively small value of c_1 in the denominator to write

$$\int_0^H \rho_0 K_{vy}\, dz = \int_0^H \frac{c_1 \rho_0 |v|^2}{2\lambda |U - c|^2} dz + O(v^3) \simeq \frac{\pi}{2} \frac{\rho_0 |v|^2}{\lambda |\partial U/\partial z|} + O(v^3) \quad . \quad (18)$$

c_1 being so small that the contribution to the integral near the steering level dominates. This formula is essentially of the same form as Eq. (17), and gives values of K_{vy} about four times as large. This is probably accounted for in terms of the fraction of its lifetime that the eddy is actually amplifying. It is interesting to note that c_1 does not now appear explicitly.

(d) *Momentum flux expressed in terms of transfer coefficients*

Since only the horizontal variation of Q_* is relevant its horizontal flux is given by

$$\overline{v \rho_0 Q_*}^x = - K_{vy} \partial \overline{(\rho_0 Q_*)}^x / \partial y,$$

and using Eq. (16) for the horizontal flux of entropy, Eq. (11) becomes

$$\frac{\partial}{\partial y}(\rho_0 \overline{uv}^x) = - \frac{f}{B} \frac{\partial}{\partial z} \left(\rho_0 K_{vy} \frac{\partial \phi}{\partial y} + \rho_0 K_{vz} \frac{\partial \phi}{\partial z} \right) + \frac{\rho_0}{B} K_{vy} \frac{\partial}{\partial y}(\rho_0 Q_*)$$

where all the terms are now understood to be zonal averages and the averaging symbol has been dropped. Some cancellation occurs when the expression (11) for $\rho_0 Q_*$ is used

$$\frac{\partial}{\partial y}(\rho_0 \overline{uv}^x) = \rho_0 K_{vy} \frac{\partial}{\partial y}(f + \zeta) - \frac{f \rho_0}{B} \frac{\partial \phi}{\partial y} \frac{\partial K_{vy}}{\partial z} - f \frac{\partial}{\partial z}(\rho_0 K_{vz}) \quad . \quad (19)$$

The first term represents the essentially-barotropic transfer of absolute vorticity, while the second and third terms are concerned with baroclinic effects associated with changes in absolute vorticity due to the shrinking and stretching of columns whose ends move along slantwise paths.

Integrating Eq. (19) from the surface $z = 0$ to the tropopause $z = H$, the last term on the right-hand side vanishes since K_{vz} vanishes at $z = 0$ (or is small, if there is a frictional sub-layer) and is to a first approximation negligible at $z = H$. Hence introducing the surface stress τ_s and assuming the stress at $z = H$ to be small in comparison

$$- \tau_s = \frac{\partial}{\partial y} \int_0^H \rho_0 \overline{uv}\, dz = \int_0^H \rho_0 K_{vy} \frac{\partial}{\partial y}(f + \zeta)\, dz - \int_0^H \frac{f \rho_0}{B} \frac{\partial \phi}{\partial y} \frac{\partial K_{vy}}{\partial z} dz \quad . \quad (20)$$

On average the divergence of the eddy flux of momentum must be positive at some latitudes and negative at others (or zero everywhere) otherwise there would be a net torque on the atmosphere. But in the troposphere $f + \zeta$ increases towards the pole practically everywhere, so with K_{vy} positive the first term gives a southward flux of vorticity, and momentum-divergence everywhere. Thus the second term must be important somewhere, most plausibly in the zone of strong horizontal temperature gradients. We then have the tentative result that westerly surface winds will tend to be found under the westerly jet.

(e) *A constraint upon K_{vy}*

The 'integrative' property of the transfer coefficients discussed in Section 8 (b), (d) allows Eq. (20) to be thrown into a rather neat form suggesting several interesting generalities. For perturbations of zonal flow in which the unperturbed flow is a function of height only it can be shown that there is no net momentum flux, so the two terms on the right-hand side of Eq. (20) cancel. Therefore if the K's in a realistic situation are the same as those for which the baroclinity and β-effect are uniform they must satisfy the relation

$$\int_0^H \rho_0 K_{vy}\, dz = - \frac{H}{\gamma} \int_0^H \rho_0 \frac{\partial K_{vy}}{\partial z} dz \quad . \quad . \quad . \quad . \quad . \quad (21)$$

where $\gamma = gB\,H^2\,\beta_1/(f^2\,\overline{\Delta U^y})$ is a parameter characteristic of the whole baroclinic zone, in which

$\overline{\Delta U^y} = U^y\,(z = H) - U^y\,(z = 0)$: the average total shear of zonal wind,

$\beta_1 = \beta - \overline{(\partial^2 U/\partial y^2)}^{yz}$: the average latitudinal variation of absolute vorticity.

The parameter γ enters the perturbation theory (as in Green 1960) as the ratio of vorticity changes due to meridional advection to vorticity changes due to vertical stretching, so in that sense barotropic compared to baroclinic influences.

9. Maintenance of surface winds : direct estimate from observation

In the troposphere $\beta - \partial^2 U/\partial y^2$ and $\partial\phi/\partial y$ vary relatively little with respect to height so can, to a fairly crude first approximation be taken outside the integral of Eq. (20) to give on use of Eq. (21),

$$\tau_s = -\left(\beta - \frac{\partial^2 U}{\partial y^2} + \frac{\gamma f}{BH}\frac{\partial\phi}{\partial y}\right)\cdot\int_0^H \rho_0\,K_{vy}\,dz. \qquad (22)$$

All these quantities can be estimated, directly or indirectly, and Fig. (9) shows the results of a calculation of each right-hand side term, using data for the winter season given by Peixoto (1960) and Buch (1954) which gives $\gamma = 1{\cdot}4$, and using $K_{vy} = 4{\cdot}10^6$ m^2 s^{-1} given by Eq. (17). The zone of positive stress should coincide with the zone of surface westerlies but agreement is not particularly impressive (contrast with Fig. 12 (a) for example), and the magnitude of the surface stress is, at 3 or 4 dyne cm^{-2}, somewhat too large. Nevertheless the comparison is encouraging in view of the extreme simplification brought about particularly through Eq. (21), and the strain placed on the data in calculating the separate terms (particularly $\partial^2 U/\partial y^2$) in Eq. (22), bearing in mind that the surface stress is considerably smaller than each term on the right-hand side.

(a) *Maintenance of surface winds : general circulation model*

The transfer theory can also be built into a model of the general circulation. Averaging the momentum equation Eq. (1) round a line of latitude, the advection term is (conventionally) separated into a vertical transfer of momentum (dominated by small-scale turbulence and expressed in terms of the zonal-mean zonal stress τ) and a horizontal transfer due to eddies of large-scale : the transfer of mean momentum by the mean flow being assumed negligible.

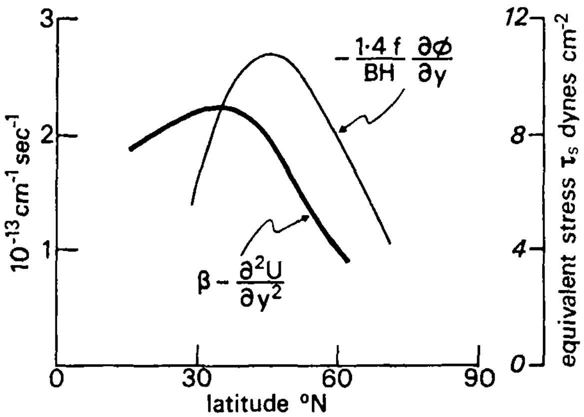

Figure 9. Latitudinal variation of terms in the momentum flux Eq. (22) contributing to the surface stress. Estimates direct from the data of Buch (1954), giving $\gamma = 1{\cdot}4$. The stress is positive where

$$-(\gamma f/BH)\,\partial\phi/\partial y > \beta - \partial^2 U/\partial y^2$$

and the transition from equatorial easterlies to mid-latitude westerlies near 30°N is realistic, but polar easterlies do not appear. Also the magnitude of the predicted stress is somewhat too large.

In order to maintain a consistent notation let (U, V, W) be the zonal-mean components of the velocity : the non-eddy component. Local time variation of the mean quantities being negligible we have

$$\frac{\partial}{\partial y}(\rho_0 \overline{uv}) - \rho_0 f V = \frac{\partial \tau}{\partial z} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (23)$$

$$\frac{\partial U}{\partial z} = -\frac{g}{f}\frac{\partial \phi}{\partial y} \quad . \quad . \quad . \quad . \quad . \quad . \quad (24)$$

Similarly for the entropy transfer defining the source function s, and continuity of mass

$$\rho_0 s \equiv \rho_0 \overline{\frac{D\phi}{Dt}} = \rho_0 V \frac{\partial \phi}{\partial y} + \rho_0 WB + \frac{\partial}{\partial y}(\rho_0 \overline{v\phi}) + \frac{\partial}{\partial z}(\rho_0 \overline{w\phi}) \quad . \quad (25)$$

and $$\frac{\partial}{\partial y}(\rho_0 V) + \frac{\partial}{\partial z}(\rho_0 W) = 0. \quad . \quad . \quad . \quad . \quad . \quad . \quad (26)$$

These equations, together with Eq. (16) and (19), form a closed set to define the seven variables

$$\overline{uv}, \quad \overline{v\phi}, \quad \overline{w\phi}, \quad U, \quad V, \quad W \quad \text{and} \quad \phi$$

assuming τ, s and the K's to be known functions of these variables. Making that reasonable assumption, solution of the set is still not quite straight-forward because of the way the surface wind U_s (say) is determined by the dynamics. Thus, though it would perhaps be more logical to take the source function s to be (more-or-less) given and deduce the flow, it is easier and equally valid to suppose ϕ given. The thermal wind equation Eq. (24) allows $U - U_s$ to be found. Then Eq. (20), with τ_s a function only of U_s, can be solved for $U_s(y)$ – the two boundary conditions requiring that the total stress, and total curl-stress should vanish at the surface. Completing the solution we find $\overline{uv}$ from Eq. (19), V from Eq. (23) then W from Eq. (26) and finally s from Eq. (25).

To the extent that this model will give the entropy transfer realistically, we know that the heating function will be related to the temperature gradient mainly through the more or less conductive process of eddy transfer, so it seems reasonable to expect that plausible profiles of ϕ should lead to plausible profiles of s, and we shall proceed on that basis. The particular advantage of the scheme is that the rather subtle and uncertain dynamical constraints can be examined in some detail without carrying through lengthy thermodynamic calculations.

(*b*) *An explicit example*

Consider a region bounded by smooth rigid walls at its northern and southern boundaries $y = \pm L$, by a smooth rigid lid at $z = H$ and a rough rigid surface at $z = 0$.

Let $$\partial\phi/\partial y = -A(L^2 - y^2) \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (27)$$

with the constant A representing the baroclinity, concentrated in middle latitudes as anticipated to be important in Section 8 (d). Eq. (24) then gives

$$U = U_s(y) + (gAz/f)(L^2 - y^2) \quad . \quad . \quad . \quad . \quad . \quad . \quad (28)$$

and Eq. (20) becomes

$$\left(\beta - \frac{\partial^2 U_s}{\partial y^2}\right)\int_0^H \rho_0 K_{vy}\, dz + \frac{2gA}{f}\int_0^H \rho_0 z K_{vy}\, dz$$
$$+ \frac{Af}{B}(L^2 - y^2)\int_0^H \rho_0 \frac{\partial K_{vy}}{\partial z}\, dz + \tau_s = 0 \quad . \quad . \quad (29)$$

The integrals involving K_{vy} are related by Eq. (21) with $\gamma = 3\,BH\,\beta_1/(2\,Af\,L^2)$, and define a pure number a through :

$$\int_0^H \rho_0\, z\, K_{vy}\, dz = aH \int_0^H \rho_0\, K_{vy}\, dz.$$

The value of a is dependent upon the β-effect but only slightly : for $\gamma = 0$, $a = 0{\cdot}5$, and for $\gamma = 1$, $a = 0{\cdot}41$. The empirical law :

$$\tau_s = \rho_s\, k\, U_s \quad \text{with } k \text{ and the surface density } \rho_s \text{ constant,}$$

gives a crude but tractable expression for the mean surface stress, using which Eq. (29) becomes :

$$\frac{d^2 U_s}{dy^2} - \mu^2 U_s = \beta_2 - \gamma \frac{Af}{BH}(L^2 - y^2) \qquad (30)$$

with
$$\mu^2 = \rho_s\, k \Big/ \int_0^H \rho_0\, K_{vy}\, dz, \quad \beta_2 = \beta + 2a\frac{gAH}{f} = \beta + \frac{3a\,\overline{\Delta U}^y}{L^2}.$$

Since the walls are rigid, v and therefore $\overline{uv}$ must vanish thereon. Thus integrating Eq. (20) with respect to y from one wall to the other, the total mean surface stress, hence $\overline{U_s}^y$, must vanish : this provides a rather powerful constraint on U_s (i.e. the mean zonal flow is unaccelerated so the total external stress must vanish). Again, differentiating the product $\overline{uv}^x$ it is evident that $\partial\, \overline{uv}^x/\partial y$ also vanishes at the walls, hence since $V = 0$ there Eq. (23) shows that $U_s = 0$ also. This turns out to be an extremely important condition.

Thus if the latitudinal variation of K_{vy} is neglected (an approximation most severe in extreme latitudes) then μ^2 is constant and Eq. (30) can be integrated analytically. The flow being symmetrical about $y = 0$ the solution satisfying $\overline{U_s}^y = 0$ is

$$U_s = \frac{\beta_1}{\mu^2} F(\mu L, y/L) - \frac{\beta_2}{\mu^2} G(\mu L, y/L)$$

where
$$F = \frac{3}{2} - \frac{3}{\mu^2 L^2} - \frac{3y^2}{2L^2} - \mu L \left(1 - \frac{3}{\mu^2 L^2}\right) \frac{\cosh \mu y}{\sinh \mu L}$$

$$G = 1 - \mu L \frac{\cosh \mu y}{\sinh \mu L} \quad \text{and} \quad \beta_1 = \beta_2 - \overline{(\partial^2 U_s/\partial y^2)}^y. \qquad (31)$$

The boundary condition $U_s(\pm L) = 0$ then determines the ratio β_1/β_2 with the following representative values :

$$\beta_1/\beta_2 = 0{\cdot}833\ (\mu L = 0),\quad 0{\cdot}849\ (\mu L = 2),\quad 1\ (\mu L \to \infty),$$

and the terms in F and G must rather closely cancel each other everywhere, just as for the corresponding observed quantities in Sections 7 and 9. U_s is positive in middle latitudes, negative in high and low – typical solutions are shown in Figs. 10 to 12.

(c) *Barotropic instability*

If the vertical shear is sufficiently small compared to the horizontal shear then this theory of baroclinic momentum-transfer must break down. Interpreting Charney and Stern (1962) as showing that practically any realistic flow will be unstable, we consider the character of the instability in extreme circumstances. If the initial wind is zonal and varies only with latitude then the growing instability must reduce the mean kinetic energy, hence transfer momentum down the gradient of initial momentum. On the other hand, in sheared flow like that shown in Fig. 1 in which the instability is baroclinic the growing eddy transfers momentum in the opposite direction. Again, to the extent that the term in G represents a barotropic effect and is about eight times as big as $|F - G|$ the momentum transfer must be about eight times as large if the flow is barotropically instead of baroclinically unstable and the eddies are of similar intensity. Thus we expect completely different behaviour, in terms of sign and intensity of momentum transfer in these two circumstances.

Now assuming that the horizontal and vertical changes of initial velocity are comparable and spread over distances L and H respectively, the condition that the available kinetic energy (the result of ' mixing ' keeping the total momentum the same) should exceed the available potential energy i.e. that barotropic instability should dominate, is roughly

$$L^2 \ll H^2 gB/f^2,$$

so flow satisfying this criterion is unlikely to be adequately treated by the theory leading to Eq. (31). Our largely subjective impression is that borderline cases will be dominated by the baroclinic nature of the instability where the shears are large, but by barotropic influences elsewhere – roughly inside and outside the surface formed by the steering levels.

(d) *Baroclinic flow with β small*

Another limiting case concerns baroclinic motion in which $\beta = 0$, and the entropy transfer and K_{vy} might be expected to be independent of height, in which case the baroclinic effect on momentum transfer would be absent. But when the instability is baroclinic the argument of Section 9 (b) giving β_1/β_2 shows that the β-effect must be significant through the curvature of the velocity profile if not through the variation of planetary vorticity. Thus for all except possibly barotropic systems, solutions like Eq. (31) are expected to be valid.

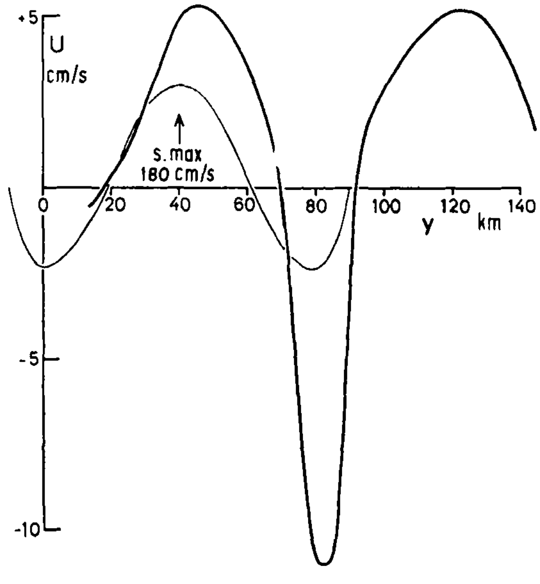

Figure 10. One 3-day average of the bottom velocities below the Gulf Stream off Cape Hatteras (thick line), and those predicted for the mean stream by Eq. (32), using contemporary values for $\overline{\Delta U}^y$ over the zone $-10 < y < 90$ km where the stream is supposed bounded by rigid walls. Note that the surface velocity attains a maximum of 180 cm s^{-1} near $y = 40$ km. The agreement between the bottom velocities is perhaps suspiciously close since the bottom velocities are nearly an instantaneous measurement compared to the eddy-lifetime. Nevertheless we are encouraged.

10. Comparison with observed systems

(a) *The Gulf Stream*

Although neither Section 9 (c) or (d) is directly applicable to the troposphere there are other rotating systems in which the Richardson number is large. Flow of the ocean through the Straits of Florida satisfies the condition that the available kinetic energy should exceed the available potential energy with $gB\,H^2/(f^2\,L^2) \simeq 6$ so any instability would be expected to induce a bottom current in the opposite sense to that at the surface. Though Pillsbury's measurements, quoted by Sverdrup, Johnson and Fleming (1942) do show such a current, later direct and more accurate measurements by Richardson, Schmitz and Niiler (1969) show no reversal. In fact the stream is probably stable here : stable to barotropic disturbances because Rayleigh's criterion (that $\beta - \partial^2 U/\partial y^2$ must change sign for barotropic instability) is not satisfied, and stable for baroclinic disturbances because the stream is too narrow to permit Eady's long baroclinic waves to develop, and the shorter waves, which grow comparatively slowly in a frictionless system, are liable to be heavily damped by bottom-friction (it is doubtful if estimates of bottom-friction are sufficiently reliable for this to be tested quantitatively).

Further along, near Cape Hatteras, the stream becomes broader and is probably baroclinically unstable : we estimate that $gB\,H^2/(f^2\,L^2) \simeq 1$ but the calculation is sensitive to the values of L and H chosen. These should represent dimensions of the stream averaged through an eddy (in time or space) and values of $2L = 100$ km (for the width) and $H = 400$ m (for the depth) of the sheared layer are not inconsistent with the data given by Stommel (1958). For values of μL not too large ($\mu L < 1$ about) the predicted bottom velocity is independent of μL and, since $\beta\,L^2 \ll \overline{\Delta U}^{\,y}$ for these dimensions, U_s can be written :

$$U_s = (1 - 6\,Y^2 + 5\,Y^4)\,\overline{\Delta U}^{\,y}/32 \qquad (32)$$

where $Y = y/L$.

Warren and Volkmann (1968) deduced bottom velocities from those observed at a depth of $2\frac{1}{2}$ km and extrapolated to the bottom at 4 km using the thermal wind equation, and these are compared with those given by Eq. (32) using $\overline{\Delta U}^{\,y} = 92$ cm s^{-1} (deduced from their simultaneous surface velocities) in Fig. 10. The agreement is very good bearing in mind the difficulty of observation and the fact that the bottom-friction is a sensitive indicator of the eddy transfer of momentum. In contrast, 140 km to the right of the main Gulf Stream there was a second velocity maximum about half as strong as the main stream and across which the baroclinity was not so systematic, associated with bottom velocities stronger than for the main stream and in the opposite sense to the local surface velocities. It is tempting to speculate that this is the result of barotropic instability.

We believe that the transfer of momentum by large-scale eddies is an important factor determining the bottom velocities and transverse circulation in the Gulf Stream, though the effect of down-stream variation should (and can) be included, and some further thought on the consequences of the variation of baroclinity with depth is desirable, but a body of data on mean bottom velocities (or bottom drag) is essential, as are estimates of (at least) μL even though these latter are unlikely to be critical.

(b) *Comparison with dishpan experiments*

The flow of liquids in heated-rotating bowls is baroclinically unstable for ranges of parameters realizable in the laboratory, but the measurement of velocity in such systems is difficult – particularly near the bottom. There is little doubt that the zonal component is reversed in the lower layers though the reported flow is often so violent as to give the impression of a net (negative) torque, which is of course unrealistic. The contribution of the side walls to the stress must be included and that of the upper (usually 'free') surface may be significant. Fig. 11 shows results reported by Fultz *et al.* (1959) and Riehl and Fultz (1957). In this Figure velocities have been normalized with respect to the mean

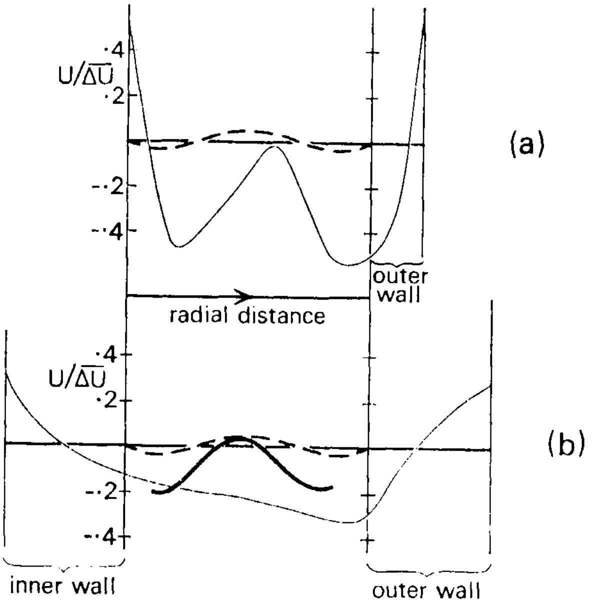

Figure 11. Bottom velocities in dishpan experiments.

(a) Composite of three 'wave-régime' experiments by Fultz using a dishpan with outer wall only. Zonal velocity measured at upper surface and extrapolated downwards using the thermal wind equation – solid line. Prediction of cartesian momentum transfer theory – broken line.

(b) Annulus experiment in 'vacillation-régime' by Riehl and Fultz. Zonal velocity extrapolated as above – thin solid line, zonal velocity inferred via Ekman-layer theory from measured meridional component – thick solid line, predicted by cartesian momentum transfer theory – broken line.

We suspect that the downward extrapolation gives rather crude estimates of the comparatively feeble bottom currents – and note that the more direct measurements are in the best agreement with the theory.

thermal wind $\overline{\Delta U}^y$ – in many respects more convenient than referring all velocities to the rim speed as is more usual in such experiments. Fig. 11 (a) represents a composite of three 'wave régime' experiments quoted by Fultz *et al.* (1959) in which the velocity has been integrated downwards using the thermal wind relation, from values observed at the upper surface. In view of several uncertainties in this procedure the rather poor agreement between theory (dashed) and experiment is perhaps not surprising. Also the theory represents the geometry of the open-centred dishpan rather poorly.

Fig. 11 (b) shows velocities, similarly deduced by Riehl and Fultz (1957) in a 'vacillation-régime' experiment. Here however, the meridional velocity near the top of the Ekman layer was also measured directly. Assuming this to coincide with the maximum transverse velocity v_f say, the Ekman-layer theory shows that the zonal velocity just outside the layer is given by $3{\cdot}1\ v_f$ and it is this value that is plotted in Fig. 11 (b). Agreement with theory is much better, even though for this experimental arrangement the effect of the rather tall walls at rim and centre (assumed frictionless in the theory) is likely to be more serious.

(*c*) *Comparison with the winter mesospheric jet*

The mesospheric winter jet is an interesting case which is probably marginal so far as barotropic-baroclinic instability is concerned with $gB\,H^2/(f^2\,L^2) \simeq 1$ for the monthly mean flow (given by $H/L = 20$ km/3,000 km), but for which the β-effect is not negligible with $\beta\,L^2/\overline{\Delta U}^y \simeq 3$ with $\overline{\Delta U}^y = 30$ m s^{-1}. If, as seems likely, there is a level between the troposphere and stratosphere where τ is small, then Eq. (22) would be a constraint

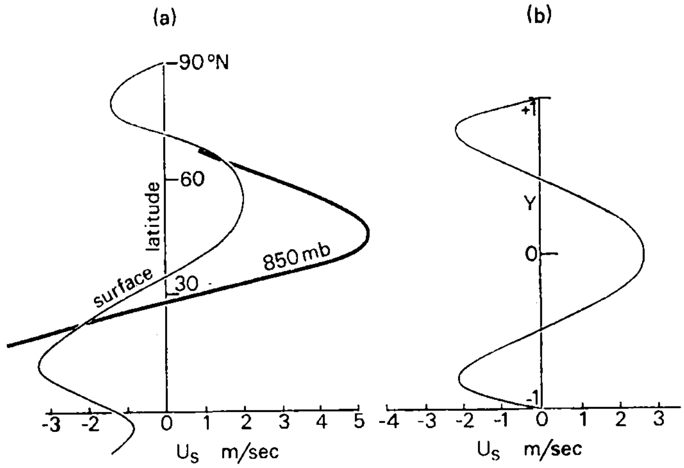

Figure 12. Variation of mean zonal component of wind with latitude in the winter season,

(a) at anemometer level (Mintz) thin, at 850 mb (Buch) thick.

(b) from theoretical solution with Y-scale such that $Y = \pm 1$ corresponds to $\pm$ 3,000 km.

upon the variation of mean velocity within the mesosphere. On a shorter time-scale a temporal evolution of mesospheric motion appears likely with the previously rejected term in $\partial U/\partial t$ becoming significant. During the polar winter, cooling generates sinking, thence meridional displacements which modify the zonal wind towards thermal-wind balance to give a global-scale jet. This, being baroclinically unstable, probably becomes concentrated into middle latitudes through baroclinic momentum transfer, and at the same time some of the polar cooling is compensated through baroclinic heat transfer. Eventually, either the summer returns or the jet, which is continually sharpened through the momentum transfer, becomes barotropically unstable. If this leads to a considerable redistribution of momentum in the horizontal, as seems likely from the considerations of Section 9 (c), then thermal-wind imbalance will again call for meridional adjustment giving a smoothing of the Pole-Equator temperature profile, warmings in high-latitudes, and coolings in low mid-latitudes of a suddenness governed by the rapidity with which the barotropic instability transfers momentum. This we believe to represent the 'sudden warming' phenomena of the late polar winters. It demands a sharp separation between barotropic and baroclinic processes that is difficult to appreciate from the theoretical point of view, though enough relevant stability problems have not yet been solved and discussed. It also seems possible to us that the sudden warming may be the last, most intense baroclinic wave of the winter, a hypothesis that might be tested by reference to the relations between eddy intensity and baroclinity outlined in Section 3.

(*d*) *Comparison with the troposphere*

The middle-latitude troposphere is baroclinically unstable – with $gBH^2/(f^2L^2) \simeq 0{\cdot}1$, and the variation of planetary vorticity is important – with $\beta L^2/\Delta U^y \simeq 10$. In this case we have enough data to estimate μ, and using $k = 1{\cdot}2$ cm s^{-1} (from observed mean winds just above the friction layer and observed mean stress), $K_{vy} = 4{\cdot}10^6$ m^2 s^{-1} and $L = 3{,}000$ km gives $\mu L = 2{\cdot}0$. Then with $\beta = 1{\cdot}5 \times 10^{-11}$ m^{-1} s^{-1} we get the values shown in Fig. 12 (b). Clearly, geometrical effects cannot be entirely neglected so the solution has been placed so that the latitude of maximum meridional temperature gradient (about 42°N) coincides with that in the model ($y = 0$) and, in addition the Y-axis has been stretched a little so that Fig. 12 (b) which is supposed to cover the whole globe is more readily comparable with Fig. 12 (a). The general shape of the variation with latitude is quite realistic, as is the magnitude of the stress ($\tau_s \simeq \pm 0{\cdot}4$ dyne cm^{-2}) particularly in contrast to the 'direct' estimates of Fig. 9. It is of some consolation that out of the three quantitative applications described here the theory best fits that with the more accurate and extensive data.

Completion of the solution is expected to be relatively straightforward, though the calculations are lengthy and the distribution of the K's discussed in Section 8 becomes important. Thus having $U(y, z)$ and $\phi(y, z)$, Eq. (19) must be solved for $\overline{uv}$ using values of K_{vy} and K_{vz}, then Eq. (23) determines V. For example, assuming that $\overline{uv}$ is negligible in a friction layer of depth z_f in which τ decreases to zero we find

$$V = kU_s/(f z_f) \quad 0 < z < z_f \quad \text{(about 30 cm s}^{-1}\text{ for } z_f = 1\text{ km).}$$

Eq. (26) can be used to find W which in this example has the magnitude :

$$W_f \simeq - z_f \, \partial V/\partial y \simeq - (k/f) \, \partial U_s/\partial y \quad \text{(about 0·024 cm s}^{-1}\text{)}$$

giving a contribution to $D\phi/Dt$ arising from the meridional circulation of $B\,W_f$, corresponding to $\pm\, 0{\cdot}08$°C day^{-1}, compared to an eddy contribution of roughly $\frac{1}{2} K_{vy}\, \partial^2 \phi/\partial y^2$ corresponding to $\pm\, 0{\cdot}5$°C day^{-1}. The net heating (radiation + convection) of the atmosphere in this model therefore varies roughly linearly with latitude and can be considered an acceptable first approximation to that observed.

11. Future developments

In addition to the points made in the text there are several lines worth developing. The geometry can be made more realistic by using spherical and cylindrical co-ordinates as appropriate, and the complete solution should be studied to discover how the spatial variation of the fluxes depends on that of the transfer coefficients. From a theoretical point of view the relation between the flux of momentum and potential vorticity is probably not so dependent on the quasi-geostrophic constraint as it has been convenient to assume here, and further discussion of the rejected terms is desirable.

It would seem reasonable to use the transfer theory for the transient eddies in a model of the global circulation, in which time-steps of the order of a month could be used and longitudinal variations included explicitly. The flux of water vapour could certainly be included crudely and the model might then be sufficiently realistic to allow useful climatic analysis and prediction.

Acknowledgments

The ideas behind this work owe much to the influence of the late Dr. E. T. Eady. I am also grateful to colleagues and to referees who helped to clarify the text.

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